Sequences and Series Cheat Sheet

Sections 9.1 – 9.6: Sequences and Series

Thomas' Calculus 15e (SI) – Summary

9.1 Sequences

Definition: $\{a_n\}$ converges to L if $\forall \epsilon > 0$, $\exists N \in \mathbb{N}$ such that $n > N \Rightarrow |a_n - L| < \epsilon$. If no such L exists, the sequence **diverges**.
Limit Laws: If $\lim a_n = A$, $\lim b_n = B$, then

$$\lim(a_n \pm b_n) = A \pm B, \quad \lim(ka_n) = kA,$$

$$\lim(a_n b_n) = AB, \quad \lim \frac{a_n}{b_n} = \frac{A}{B} \ (B \neq 0).$$

Sandwich Theorem: If $a_n \leq b_n \leq c_n$ and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.
Continuous Function Theorem: If $a_n \to L$ and f is continuous at L, then $f(a_n) \to f(L)$.
L'Hôpital's Rule for Sequences: If $a_n = f(n)$ and $\lim_{x\to\infty} f(x) = L$, then $\lim a_n = L$.
Common Limits:

$$\frac{\ln n}{n} \to 0, \quad n^{1/n} \to 1, \quad x^{1/n} \to 1 \ (x > 0), \quad \left(1 + \frac{x}{n}\right)^n \to e^x$$

Bounded Monotonic Sequences: Every bounded monotonic (increasing or decreasing) sequence converges.

9.2 Infinite Series

Definition: $\displaystyle\sum_{n=1}^{\infty} a_n$; partial sums $s_n = \sum_{k=1}^{n} a_k$. If $\lim s_n = L$ exists, the series converges to L (its sum).

Geometric Series: $\displaystyle\sum_{n=1}^{\infty} ar^{n-1}$

- Converges to $\dfrac{a}{1-r}$ if $|r| < 1$.
- Diverges if $|r| \geq 1$.

nth-Term Test:

- If $\sum a_n$ converges, then $a_n \to 0$.
- The converse is false (e.g., harmonic series).
- If $a_n \not\to 0$, the series diverges.

Algebraic Rules: $\sum(a_n \pm b_n) = \sum a_n \pm \sum b_n$, $\sum ka_n = k\sum a_n$ (if both series converge). Adding or deleting finitely many terms does not affect convergence.

9.3 Integral Test and p-Series

Integral Test: Let $a_n = f(n)$ with f positive, continuous, and decreasing for $x \geq N$. Then

$$\sum_{n=N}^{\infty} a_n \quad \text{and} \quad \int_N^{\infty} f(x)\,dx$$

either both converge or both diverge.

p-Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$\begin{cases} p > 1 & \text{converges} \\ p \leq 1 & \text{diverges} \end{cases}$$

Examples:

$$\sum \frac{1}{n} \text{ diverges}, \quad \sum \frac{1}{n^2} \text{ converges}, \quad \sum \frac{1}{\sqrt{n}} \text{ diverges}.$$

9.4 Comparison Tests

Direct Comparison: Assume $0 \leq a_n \leq b_n$ for all $n \geq N$.

- If $\sum b_n$ converges, then $\sum a_n$ converges.
- If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Limit Comparison: Let $a_n, b_n > 0$ and $L = \lim_{n\to\infty} \dfrac{a_n}{b_n}$.

- If $0 < L < \infty$, both series converge or both diverge.
- If $L = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
- If $L = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

9.5 Absolute Convergence, Ratio and Root Tests

Absolute Convergence: If $\sum |a_n|$ converges, then $\sum a_n$ converges. (The converse is false.)

Ratio Test: $\rho = \lim_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right|$
$\rho < 1 \Rightarrow$ absolutely convergent
$\rho > 1$ (or ∞) $\Rightarrow$ divergent
$\rho = 1 \Rightarrow$ inconclusive.
Root Test: $\rho = \lim_{n\to\infty} \sqrt[n]{|a_n|}$; same criteria as Ratio Test.

9.6 Alternating Series and Conditional Convergence

Alternating Series: $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} u_n$, $u_n > 0$.

Alternating Series Test: If $\{u_n\}$ is decreasing and $u_n \to 0$, the series converges.
Alternating Series Estimation Theorem: For a convergent alternating series,

$$|R_n| = |L - s_n| \leq u_{n+1},$$

and R_n has the same sign as the first omitted term.
Conditional Convergence: A series that converges but does not converge absolutely is called conditionally convergent.
Example: The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

converges (its sum is $\ln 2$), but it is not absolutely convergent (the harmonic series diverges).

Quick Tips:

- For series with factorials or powers, Ratio or Root Test often works.
- For series resembling $1/n^p$, use p-series or comparison.
- Alternating series: always check $u_n \to 0$ and decreasing.
- The nth-term test only detects divergence, not convergence.

Strategy for Series

from Paul's Online Notes

https://tutorial.math.lamar.edu/classes/calcii/seriesstrategy.aspx

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Guidelines for Determining Convergence

1. **Divergence Test:** Quick glance: $\lim_{n\to\infty} a_n \neq 0$? If so, the series diverges.
2. **p-series / Geometric:** Is it $\sum 1/n^p$ or $\sum ar^n$?
   - p-series converges if $p > 1$, diverges if $p \leq 1$
   - Geometric series converges if $|r| < 1$, diverges if $|r| \geq 1$
3. **Comparison Test:** Similar to p-series or geometric? Try Comparison Test.
4. **Rational Expressions:** If the series involves only polynomials (or polynomials under radicals), try Comparison Test or Limit Comparison Test. (Requires positive terms.)
5. **Ratio Test:** Contains factorials or constants raised to n? Ratio Test often works.
6. **Alternating Series Test:** Can terms be written as $a_n = (-1)^n b_n$ or $a_n = (-1)^{n+1} b_n$? Try Alternating Series Test.
7. **Root Test:** Can terms be written as $a_n = (b_n)^n$? Try Root Test.
8. **Integral Test:** If $a_n = f(n)$ with f positive, decreasing, and $\int f(x)\,dx$ is easy, try Integral Test.

Important Note: More than one test may apply. Go through all guidelines and choose the easiest test for you.

References

- Thomas, G. B., Weir, M. D., & Hass, J. (2019). *Thomas' Calculus* (15th ed.). Pearson.
- Dawkins, P. (2022). Strategy for series. *Paul's Online Notes*. Retrieved from https://tutorial.math.lamar.edu/classes/calcii/seriesstrategy.aspx